

UXL SKILLS EVALUATOR

Performance Validation Skill

Correctness-first benchmark and claim discipline across UXL projects

September 2026 • 4 of 6 evaluation tasks implemented

Why this is useful across UXL projects

- Stop premature speedup claims — Correctness and a valid baseline must pass before performance is discussed.
- Make timing comparable — Setup, transfer, warmup, steady state, synchronization, topology, and resource limits are declared.
- Handle numerical reality — Tolerances and non-associative reductions are part of the test contract.
- Keep claims bounded — Results describe the measured software, model, harness, toolchain, and hardware cell—not a universal score.

What the skill tells an agent today

- Write the user-visible correctness contract first — Outputs, metrics, tolerances, and failure criteria.
- Choose the baseline and scope — State exactly which work each timing includes.
- Warm up and repeat — Report distribution and variance, not a single best run.
- Synchronize asynchronous work — Host timers stop only after the measured operation completes.
- Profile after a validated regression — Profilers explain a measured problem; they do not create one.

Evaluation inventory: 4 implemented, 2 target gaps

- **Implemented** — Tiny async GPU claim; benchmark report repair; floating-reduction tolerance; cgroup concurrency quota.
- **Incident evidence** — The concurrency task is grounded in a oneTBB maintainer incident.
- **Planned** — Transfer-inclusive comparison and profile-after-regression on declared target lanes.
- **What this proves** — Current tasks test evidence discipline; none yet retains measured skill headroom for promotion.

How we chose the content

- **Project-native benchmarks first** — Each library's tests and benchmark tools remain authoritative.
- **Cross-project invariants** — Correctness, baseline, scope, synchronization, variance, provenance, and claim language recur everywhere.
- **Adversarial scenarios** — Tasks target tempting but invalid conclusions, not just command recall.
- **Visible gaps** — Target-dependent measurement remains planned until qualified lanes and authentic regressions exist.

What shared ownership would look like

- Working-group reviewers own — The common evidence and claim contract.
- Project benchmark owners supply — Approved commands, metrics, tolerances, representative sizes, and interpretation limits.
- Hardware owners qualify lanes — They prove the environment, not the skill's value.
- UXL infrastructure maintains — Matched evaluation cells, provenance schemas, Harbor artifacts, and public dashboards.

A focused 30-minute maintainer review

- 1. Confirm the evidence order — Correctness, baseline, scope, repetitions, variance, then profiling.
- 2. Correct one benchmark assumption — Add a project-specific limitation or required metric.
- 3. Judge the six tasks — Keep, rewrite, or remove; nominate an authentic regression with headroom.
- 4. Name a shared owner — Include project benchmark owners in periodic review.
- Outcome — Agents produce reproducible, narrowly scoped evidence before making performance claims.